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The Citizen Data Scientist at Cosecha Fresca

Camila Restrepo,^a Chief Product Officer at Cosecha Fresca, a coffee distribution and grocery sales company, gazed at the framed photograph of Finca El Cafetal's rolling hills and coffee plantations on her desk. It was a reminder of how far Cosecha Fresca had come from its humble beginnings, and looking at it always grounded her before an important meeting. As 2025 drew to a close, she knew the conversation she would have with her leadership team would shape the company's future in profound ways.

The question on the table was both complex and straightforward: Could Restrepo herself, leveraging the new generation of generative AI (GenAI) tools, tackle the data science challenges facing Cosecha Fresca? Or did the company need to hire dedicated data scientists with formal training in statistics and machine learning (ML)? Hiring a data science team would require substantial investment—senior data scientists in Colombia commanded salaries well over 200 million pesos^b annually, and building even a small team would represent a significant commitment. Outsourcing to consultants was another option, but the company's experience with consultants had been mixed. Although the forecasting algorithm developed by consultants for Cosecha Fresca performed well, the company's internal team had a limited understanding of its operation, making it difficult to modify or improve.

The landscape of data science changed dramatically in 2023 and 2024 with the emergence of powerful GenAI tools that promised to “democratize” data science, claiming to allow anyone to perform sophisticated statistical analyses without advanced technical knowledge or training. Restrepo had read about “citizen data scientists”—professionals who leveraged modern AI platforms to extract insights from data without formal training in statistics or ML. What if Cosecha Fresca could get much of the value of data science without the full cost and complexity of building a dedicated team?

During a recent company leadership meeting, several opportunities emerged. The current forecasting algorithm was excellent at predicting aggregate demand but struggled with product-level predictions. Cosecha Fresca's customer base had become increasingly diverse, ranging from loyal weekly subscribers to occasional one-time purchasers. The company wanted to create customer

^a See Exhibit 2 for a Restrepo's background.

^b 200 million Colombian pesos (COP) is equivalent \$53,500 as of 12/31/2025

Professor Iavor I. Bojinov and Lecturer William H. Marks prepared this case. This case is not based on a single individual or company but is a composite based on the author's general knowledge and experience. Funding for the development of this case was provided by Harvard Business School. HBS cases are developed solely as the basis for class discussion. Cases are not intended to serve as endorsements, sources of primary data, or illustrations of effective or ineffective management.

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segments to better understand needs and desires. Next, there was a growing desire to optimize marketing efforts for the company's premium products. Finally, the team sought to improve the existing forecasting algorithm^c to create enhanced demand forecasts for the company's Finca El Cafetal Specialty Beans. However, Restrepo reflected on a warning provided by Santiago Vargas, Cosecha Fresca's lead user experience designer - "Whatever we do, it can't be a black box. We need to understand our customers better, not just target them with algorithms."

"What if I were to take this on myself?" Restrepo suggested to the rest of the company leadership. "Not as a data scientist, but as a 'citizen data scientist.' I want to see if I can tackle these challenges directly using the new AI tools like Julius.ai, Manus, Gemini, or ChatGPT—to understand what is possible, what the limitations are, and whether this approach can work for Cosecha Fresca before we commit to hiring a full team or going back to consultants."

The proposal sparked immediate discussion. It was an appealing low-risk experiment that would help Cosecha Fresca understand when it truly needed professional data scientists versus when business leaders could handle the work themselves. However, there were concerns about whether Restrepo had the technical foundation to evaluate the quality and trustworthiness of AI-generated models.

After considerable discussion, the leadership team agreed. Restrepo would dedicate significant time over the next several weeks to exploring what was possible and determining whether citizen data science could play a meaningful role in Cosecha Fresca's future. The team identified three specific projects as test cases: customer segmentation, premium product marketing optimization, and enhanced individual product demand forecasting.

Cosecha Fresca

Beginnings of a Farm Product Platform

Cosecha Fresca was founded in 2008 as a website for Finca El Cafetal in the coffee-growing region of Antioquia, Colombia (Exhibit 1A), to sell its coffee and related products to customers who could not visit the farm or its stands. In 2014, Cosecha Fresca partnered with a dairy farm in Cundinamarca (Exhibit 1B) and a vegetable farm in Boyacá (Exhibit 1C) to increase its product offerings. See Exhibit 3 for screenshots of the company's website. The partnership's success led the Finca El Cafetal leadership team to spin off Cosecha Fresca as an independent company in 2016 to simplify partnering with new farms and accelerate growth.

By 2019, Cosecha Fresca had partnered with seven farms across the Andean region and launched a weekly grocery box subscription service that delivered seasonal produce. The new box service enabled Cosecha Fresca to produce a more accurate demand forecast, thereby drastically reducing food losses from storage. "The box service is our path to a steady recurring revenue stream that will enable us to scale our operations while considerably reducing our number one cost—food perishing," Restrepo said at the time.

Between 2019 and 2020, Cosecha Fresca focused heavily on developing its logistics and technology capabilities to ensure the weekly grocery box subscription could scale effectively. Leveraging its partnerships with several farms, the company established a regional distribution network connecting Bogotá, Medellín, and Cali, enabling faster fulfillment and fresher deliveries. Cosecha Fresca also began

^c See Exhibit 10 for a glossary of terms throughout the case.

investing in digital marketing, highlighting its direct-to-consumer convenience and fresh, locally sourced produce at a time when online grocery shopping was on the rise in Colombia's major cities.

Expanding Capabilities

In 2021, Restrepo hired external consultants to develop a sophisticated forecasting algorithm that leveraged historical sales data, seasonal trends, and partner farm output. This model refined the company's demand planning, further reducing food waste and strengthening relationships with growers. The company significantly benefited from the COVID-19 pandemic, which led to rapid growth in demand for online grocery ordering. Cosecha Fresca opened a second distribution hub in Medellín to meet growing demand, prioritizing temperature-controlled storage to maintain peak freshness for fruit, vegetables, dairy products, and coffee.

Starting in 2022, the company placed renewed emphasis on forming strategic partnerships with like-minded organizations, including small-scale producers of pantry staples like artisanal masarepa, panela, and bocadillo.^d Cosecha Fresca integrated these new offerings into the weekly box delivery platform, creating a more comprehensive grocery experience for its subscribers. The expanded lineup also helped Cosecha Fresca appeal to a broader demographic of consumers seeking convenient access to local foods.

Over the course of 2023, Cosecha Fresca accelerated investments in technology and data analytics. The leadership team introduced a predictive inventory management system designed to track product quality in real-time—from harvest at partner farms to storage and shipment at distribution facilities. This system enabled a more flexible approach to supply, allowing the company to quickly reallocate inventory in response to changes in demand or unexpected surpluses from partner farms.

By 2024, Cosecha Fresca had firmly established itself as a leading direct-from-farm grocery delivery business in Colombia, with revenues reaching 90 billion pesos.^e Its subscriber base had grown steadily, bolstered by the platform's reputation for high-quality products and strong community ties to local agriculture. The company was on track for continued growth in 2025, with revenues expected to exceed 110 billion pesos;^f see Exhibit 4 for the company's financials and Exhibit 5 for its most popular products. Internally, the team was proud of how far they had come, yet they recognized that continued success would depend on strategic choices in customer engagement, technological innovation, and data capabilities—areas they believed would define Cosecha Fresca's next chapter of growth.

Despite this success, Cosecha Fresca had never built a dedicated data science team. The forecasting algorithm that had served them so well since 2021 had been developed by external consultants and was maintained by the engineering team. While it worked effectively for aggregate demand prediction, the company lacked the in-house expertise to extend it to more sophisticated applications or to tackle new data science challenges. This gap had become increasingly apparent as competitors began deploying personalized recommendations, dynamic pricing, and advanced customer analytics.

^d Masarepa is the corn flour used to make arepas, corn pancakes that can be sweetened or unsweetened and served with condiments or like a sandwich. Panela is a traditional, unrefined whole cane sugar often sold in solid blocks or cones. Bocadillo is a shelf-stable guava paste. All three are commonly found in Colombia.

^e 90 billion COP is equivalent to \$24 million as of 12/31/24

^f 110 billion COP is equivalent to \$30 million as of 12/31/25

Understanding the Data Science Project Lifecycle

Before delving into the three specific projects, Restrepo conducted research on how professional data scientists approached their work. She wanted to understand the structured process underlying successful data science projects, not just to follow it herself, but to better evaluate what she would need to do and where she might encounter challenges.

Through her research and conversations with colleagues at other companies, Restrepo learned that successful data science projects typically traversed five critical steps: selection, development, evaluation, adoption, and management¹. Understanding this lifecycle would be essential for her experiment.

Selection Phase

The selection phase involved assessing both the impact and feasibility of potential projects. For impact, Restrepo needed to consider strategic alignment (did the project support Cosecha Fresca's goals?), measurable outcomes (could she define clear success metrics?), and whether the AI would augment or replace human decision-making. For feasibility, she needed to evaluate whether the problem was suitable for AI, whether she had access to the necessary data, whether she had the technological capabilities to implement solutions, and what ethical considerations might arise around bias, privacy, and transparency.

Restrepo had already completed much of the selection work by identifying her three projects. Customer segmentation aligned with Cosecha Fresca's goal of better understanding and serving its diverse customer base. Premium product recommendations could directly increase revenue by helping customers discover high-margin items they would love. Enhanced demand forecasting would reduce waste and improve relationships with partner farms. All three projects appeared feasible: she had years of transaction data, the problems seemed well-suited to ML, and the ethical risks appeared manageable since the applications would primarily be internal or would augment rather than replace human judgment.

Development Phase

The development phase was where Restrepo expected to spend most of her time. Professional data scientists conducted multiple iterations of data discovery and cleaning, exploratory analysis, model training, and performance evaluation; see Exhibit 6 for definitions of the main classes of ML models. This was not a linear process but rather an iterative cycle of experimentation and refinement. Restrepo had access to modern AI platforms that could automate parts of this process, but she understood that her business knowledge would be essential for interpreting results and guiding the analysis in meaningful directions. She would need to formulate clear questions, provide context regarding Cosecha Fresca's business, and critically evaluate whether the AI's outputs were reasonable. However, given her lack of training and knowledge in statistics or ML, Restrepo knew she would be limited in understanding how and why the models worked (or did not).

Evaluation Phase

The evaluation phase would be critical before any deployment. Restrepo would need to rigorously test each model's performance against objective metrics. For the forecasting model, this would mean comparing predictions against actual demand over several weeks. For the recommendation engine, it would require measuring whether recommended products actually led to increased purchases. For customer segmentation, evaluation entailed validating that the identified segments exhibited distinct

behaviors and responded differently to marketing efforts. Restrepo recognized that this phase would be challenging for a citizen data scientist. She lacked formal training in statistical evaluation methods and wasn't sure how to assess whether a model's performance was "good enough" for real-world use.

Adoption Phase

The adoption phase would present perhaps the biggest organizational challenge. Getting people to actually use the insights and tools she developed would require more than just technical success. It would demand change management, training, and building trust in the models' outputs. Restrepo understood that adoption would be easier for internal-facing applications like segmentation and forecasting, where she could work directly with her colleagues to integrate the insights into their workflows. Customer-facing applications like recommendations would require much more careful consideration.

Management Phase

Finally, the management phase would ensure ongoing success. AI models were not "set it and forget it" solutions. They needed monitoring, updating, and maintenance as customer behaviors evolved, product offerings changed, and new data became available. This long-term management would require dedicated resources and processes that Cosecha Fresca did not yet have in place. Restrepo knew that even if she successfully built models, sustaining them over time would be a distinct challenge.

Understanding this lifecycle helped Restrepo recognize the full scope of her undertaking. She had a realistic sense of where she might succeed and where she would likely encounter limitations.

Understanding the Basics and Applications of Machine Learning

As Restrepo continued her efforts to become a citizen data scientist, she searched online to learn the basics of ML, hoping to better understand techniques for data cleaning, model development, and evaluation. In the process of training software, called the model, to predict outcomes or generate content, she learned that there were two primary types of models based on how they "learn" and make predictions: supervised and unsupervised learning. Reinforcement learning and GenAI are two other types of models that she decided not to pursue.

Supervised Learning

Supervised learning models functioned by analyzing "labeled data"—datasets that already contained the correct answers—to identify the mathematical relationships between input features and target outcomes. The process was "supervised" because these known answers guided the model's training, much like a teacher supervising a student. Restrepo likened this to preparing for a standardized test: by practicing with questions that included an answer key, she internalized the logic needed to solve new problems on the actual exam. In business applications, supervised learning primarily addressed two types of tasks: regression and classification. Regression models predicted continuous numerical values, such as the inventory quantities required for subscription boxes. In contrast, classification models assigned inputs to distinct categories, such as predicting customer churn. Exhibit 7 includes potential algorithms Restrepo found.

Unsupervised Learning

In contrast, unsupervised learning models analyzed raw data without labels or “correct” answers. Because the system operated without an “answer key” or human guidance to verify outcomes, the process was termed “unsupervised.” Rather than mapping inputs to a known output, the model’s objective was to discover hidden structures and intrinsic patterns within the dataset. The most widely applied technique was clustering, which organized data points into groups based on similarity. This approach differed fundamentally from classification: while classification sorted items into predefined categories, clustering revealed the natural groupings within the data, thereby identifying categories for the user. Exhibit 8 includes potential algorithms Restrepo found.

Hybrid

Hybrid models combined multiple algorithms from supervised and unsupervised learning to provide more powerful, more efficient, or more effective performance. These models often used different sequential algorithms to mix and match the strengths and weaknesses of different approaches. The most popular methods were self-supervised learning methods that formed the foundations of training large language models, in which the algorithm was trained on vast amounts of unlabeled data to predict a hidden segment of the input from the remaining context, such as predicting the next word in a sentence. By generating its own supervisory signals from the data itself, the model learned rich, generalized representations of language and logic at a massive scale without incurring the prohibitive costs and operational bottlenecks of human labeling. These tended to be more complex and more involved and were likely beyond Restrepo’s capabilities, but she looked into them, nonetheless. Exhibit 9 includes potential algorithms Restrepo found.

Data

Data was generally categorized into two primary forms: structured and unstructured. Restrepo visualized structured data as a traditional spreadsheet, where information was rigorously organized into rows and columns. In this format, features were the columns in the spreadsheet, each containing clearly defined numerical or categorical values—such as sales figures or dates—that algorithms could easily ingest. Conversely, unstructured data represented information without a pre-defined organization, like free-form meeting notes, call recordings, and promotional images. Unlike rows in a database, this complex dataset required more sophisticated processing to analyze the messy, human-generated content and extract the patterns necessary for prediction.

Exploratory Data Analysis

Before diving headfirst into models and using ML algorithms, Restrepo read that she should perform exploratory data analysis to better understand the data with which she was working. She would be able to understand the shape of the data, the various parameters and features, and what data was missing or needed to be cleaned. In part, she could understand what variety of options existed for each entry, how to fix, and how to prepare data for use. By systematically examining the dataset through summary statistics and visualizations—such as histograms, scatter plots, and correlation matrices—Restrepo could uncover the underlying structure of the information and identify potential quality issues. This process served as a critical safeguard, ensuring that any subsequent modeling would be based on a verified and reliable understanding of the ground truth rather than blind assumptions.

With the data science project framework and a basic understanding of ML models in mind, Restrepo was ready to begin her three projects.

Project 1: Customer Segmentation

Understanding the different types of customers Cosecha Fresca served seemed like a foundational step. Restrepo's objective was clear: to move beyond simple distinctions like "subscriber" versus "guest" and identify more nuanced groupings based on actual purchasing behavior.

The Task

Restrepo needed to determine the most effective learning approach and algorithm to identify groups of customers with similar characteristics and behaviors. These segments could then inform targeted marketing strategies, product development priorities, and customer experience improvements.

Why This Project?

This was an ideal starting point for several reasons. First, it was relatively low risk, as the output would be used for internal analysis and strategic planning rather than directly affecting customers. Second, it leveraged one of Cosecha Fresca's key data assets: years of transaction history. Third, the insights could have immediate practical value for the marketing and product teams. Finally, it was a project in which Restrepo's deep knowledge of Cosecha Fresca's customers and business model would be invaluable for interpreting the results.

Available Data

Restrepo had access to rich data from Cosecha Fresca's sales and customer relationship management systems. This included purchase histories showing which products each customer had purchased and when, order frequency and basket size, subscription status (weekly subscriber versus one-time purchaser), and products purchased.

Getting Started

Restrepo planned to export the relevant data from Cosecha Fresca's systems into a format suitable for analysis. She would need to clean the data, addressing issues such as missing values, inconsistent formatting, and outliers. Then she would use an AI platform that allowed her to upload the data and use natural language prompts to perform clustering analysis without needing to learn to write code.

Questions to Explore with AI

Restrepo prepared a list of prompts she could use to guide the AI platform, but wasn't sure if they could be improved:

- "Perform some exploratory data analysis and share the key visualizations and takeaways from the data."
- Analyze this customer transaction data and identify distinct customer segments based on purchasing behavior. What are the key characteristics that differentiate each group?"
- "What are the primary differences in purchasing habits between our weekly subscription customers and those who only make one-time purchases? Are there sub-segments within each group? Do customers convert from one-time purchases to weekly subscribers?"

- “Create a clustering model to group customers by their product preferences and shopping patterns.”
- “For each customer segment you identify, provide a profile including typical basket size, order frequency, preferred product categories, and price sensitivity.”
- “Which customer segments represent the highest lifetime value for Cosecha Fresca? Which segments have the most growth potential?”

Expected Challenges

Restrepo anticipated several challenges. First, she would need to determine the optimal number of customer segments—too few, and the insights would be too general; too many, and they would be difficult to act upon. Second, she would need to interpret the segments in a business context, translating statistical patterns into meaningful customer archetypes. Third, she would need to validate that the segments were stable and actionable rather than merely statistical artifacts. Finally, she would need to assess whether she understood the assumptions underlying the clustering algorithm well enough to trust its outputs.

Project 2: Targeting the Right Customer

For her second project, Restrepo planned to tackle a more targeted goal: optimizing marketing efforts for the company’s premium products. The objective was to move beyond broad segmentation and identify specific customers likely to upgrade their purchase habits, thereby increasing the average order value without wasting marketing budget.

The Task

Restrepo decided to focus on Cosecha Fresca’s flagship product, the *Finca El Cafetal Specialty Beans*. Rather than building a system that suggested any product to any customer, the goal was to build a prediction model. The system would need to analyze customer profiles to determine the likelihood of a specific customer purchasing this premium item in the near future. This would allow the marketing team to send exclusive offers only to those most likely to convert, rather than spamming the entire customer base.

Why This Project?

Personalized targeting is essential for high-margin, niche products. While staples like eggs and milk sold themselves, premium coffee required the right customer fit. Sending a discount code to the entire database was inefficient and risked diluting the brand’s premium value. By identifying “hidden” high-potential customers, those who fit the profile of a specialty coffee drinker but hadn’t made the leap yet, Restrepo hoped to increase revenue efficiently using data rather than intuition.

Available Data

Restrepo would use the same customer dataset from the segmentation project, leveraging the work she had already done. This included data on total engagement, spending habits, and the customer’s “share of wallet” dedicated to coffee versus groceries. Crucially, she set aside a small “holdout” group of customers. The model would remain blind to their actual behavior during training, allowing Restrepo to test if the AI could accurately predict the real-world decisions of these specific individuals.

Getting Started

Restrepo planned to start by exploring the data to understand what distinguished a “premium coffee buyer” from a standard shopper. Did they spend more overall? Did they frequently buy complementary items like chocolate or artisanal cheese? Once she understood the patterns, she would use the AI platform to predict which customers in her holdout group were likely to buy the beans.

Questions to Explore with AI

Restrepo prepared prompts to guide the development of the recommendation system, but wasn’t sure if they could be improved:

- “Analyze the dataset to identify the characteristics of customers who buy *Finca El Cafetal Specialty Beans*. Do they differ significantly from those who only buy standard coffee?”
- “Build a model to predict the likelihood of a customer purchasing the specialty beans. Which variables are the strongest predictors?”
- “I have a holdout list of 20 customers. Based on your model, predict which of these customers likely purchased the beans and which did not.”
- “Compare your predictions to the actual outcomes for the holdout group. Where did the model succeed, and where did it fail?”
- “If we have a limited marketing budget and can only send coupons to the top 10% of customers, who should we target to maximize our return?”

Expected Challenges

Restrepo anticipated a few hurdles in using AI for this task. First was the “Accuracy Paradox.” Since premium coffee was a niche product, most customers *won’t* buy it. A model could achieve 90% accuracy simply by predicting that *no one* will buy it, but that would be useless for a marketing campaign. She needed to ensure the AI focused on identifying the few positive cases (Recall) rather than just maximizing overall accuracy.

Second, she worried about interpretability. If the AI correctly identified the 20 holdout customers, could it explain *why*? A “black box” prediction (“Customer A is a buyer”) was less useful for long-term strategy than understanding the drivers (“Customer A is a buyer because their average order value exceeds \$60k”).

Finally, there was the risk of overfitting. She needed to ensure that the AI didn’t merely memorize the specific quirks of the training data (e.g., a particular random date) but instead learned generalizable patterns that would apply to future customers.

Project 3: Enhancing Demand Forecasting

For her final project, Restrepo returned to the operational heart of the business. She decided to focus on predicting demand for the Finca El Cafetal Specialty Beans, the company’s most expensive and prestigious product.

The Task

Restrepo’s goal was to use AI to generate a daily demand forecast for this specific product for the upcoming month. Unlike the stable demand for staples like eggs, sales for specialty coffee were erratic,

spiking during promotions and fluctuating with the calendar. She needed a model that could anticipate these movements better than a simple average.

Why This Project?

Forecasting this specific item was critical for cash flow. The specialty beans were expensive to hold in inventory. Overstocking tied up capital and risked spoilage, while understocking meant missed revenue and disappointed VIP customers. A precise forecast would allow the operations team to optimize inventory levels, minimizing waste and maximizing profit. If she could build a good model for one of the products, she could extend it to the rest of the inventory.

Available Data

Restrepo prepared a single dataset containing two years of daily sales history. The data included the quantity sold, the unit price (which changed during sales), and key calendar markers like holidays and paydays. Crucially, she held back the most recent month of data. She planned to hide this from the AI, generate a forecast, and then compare the prediction against what actually happened.

Getting Started

Restrepo planned to treat the AI as a data analyst. First, she would ask it to visualize the history: Was there a growth trend? Did sales peak on weekends? Then, she would ask it to quantify the impact of her business levers; specifically, how much lift did a promotion actually generate? Finally, she would ask for the prediction.

Questions to Explore with AI

Restrepo prepared prompts for developing the forecasting model, but wasn't sure if they could be improved:

- "Analyze the sales history. Describe the overall trend and any recurring seasonal patterns."
- "How much price elasticity does this product have? Calculate the average lift in sales when a promotion is active."
- "Identify outliers in the data. Are there days when sales drop to zero? Should we treat these as anomalies?"
- "Train a forecasting model to predict daily sales for the next 30 days, accounting for upcoming holidays."
- "Compare your forecast against the actual holdout data. Did the model successfully predict the peaks and valleys?"

Expected Challenges

Restrepo anticipated that this would be the "messiest" project. Real-world sales data is rarely clean; she knew there were past events (like stockouts or viral social media posts) that weren't explicitly labeled in the data. She worried the AI might misinterpret these anomalies as permanent trends. She also faced the "Black Box" problem: if the AI predicted a massive surge for next Tuesday, would she trust it enough to order double the inventory?

The Questions Ahead

As Restrepo prepared to begin her work, she found herself considering the deeper questions her experiment would need to address. The central question, could a citizen data scientist really do this work, was only the beginning. Equally important was whether she should do this work, and under what circumstances.

The new generation of AI tools had dramatically lowered the technical barriers to data science. Restrepo could certainly use these platforms to build models, generate predictions, and extract patterns from data. But could she do it well? Could she evaluate the quality of the models, understand their assumptions, understand their inner-workings, assess their limitations, and determine when they were trustworthy enough for real-world deployment? She had deep business knowledge but lacked formal training in statistics and machine learning. Would that domain expertise be sufficient, or would critical gaps in her technical understanding lead to flawed conclusions or dangerous deployments? Would the models hold up over time as the customer base evolved, or would the company end up in a similar position in the near future?

Moreover, Restrepo recognized potential gaps in her expertise. She couldn't fully assess potential biases in her training data or evaluate whether the models would perform equitably across different customer segments. She had no established framework for ensuring the models' outputs remained accurate as customer preferences shifted or as Cosecha Fresca's product mix evolved. Would attempting to do this work herself actually slow down Cosecha Fresca's progress compared to hiring professionals? Would it create technical debt that would be costly to unwind later?

As she looked once more at the photograph of Finca El Cafetal, Restrepo felt both excitement and trepidation. She was about to embark on a journey that would test not just her analytical skills, but also her judgment about when to trust AI tools and when to seek expert help. The experiment would provide valuable insights regardless of the outcome. If she succeeded in generating actionable insights, it would demonstrate the potential of citizen data science. If she encountered insurmountable challenges, that too would be valuable information, clarifying where professional data scientists were truly needed.

Exhibit 1 Departments of Colombia

A: Map of Antioquia, a department of Colombia, home to a premier coffee-growing region; B: Map of Cundinamarca, a department of Colombia, home to the dairy farm; C: Map of Boyacá, a department of Colombia, home to the vegetable farm



Source: Casewriters (Map data ©2026 Google).

Exhibit 2 Camila Restrepo Biography

Camila Restrepo was the Chief Product Officer of Cosecha Fresca. She earned a dual degree in Marketing and Business Administration from the *Escuela Superior de Negocios del Eje Cafetero*, a private business school in Pereira with close ties to Colombia's agricultural and consumer goods sectors. During her studies, Restrepo focused on brand strategy, consumer behavior, and go-to-market planning, developing an early interest in how storytelling, trust, and transparency shape purchasing decisions, particularly for food and beverage products.

Restrepo began her career in brand and growth roles at Sabores del Origen, a Medellín-based consumer goods company specializing in packaged coffee and pantry staples, where she worked on customer acquisition and retail partnerships. She later joined CompraFácil Colombia, an emerging online grocery platform, as a growth marketing manager, leading campaigns to improve customer retention. Prior to Cosecha Fresca, she served as Head of Customer Strategy at Raíz Urbana, a subscription-based food delivery startup, where she collaborated closely with product and design teams despite not having a technical background herself. Outside of work, Restrepo was a devoted coffee lover who regularly visited coffee farms across Antioquia and Caldas, cultivating relationships with growers and staying deeply engaged with specialty coffee culture—an affinity that ultimately drew her to Cosecha Fresca and informed her conviction that technology should enhance, not obscure, the human story behind the product.

Source: Casewriters

Exhibit 3 Screenshots from Cosecha Fresca's Website

**100% Organic & Local**

Sourced directly from our partner farms in Antioquia, Cundinamarca, and Boyacá.

**Farm to Door in 24h**

Our streamlined logistics ensure your produce arrives at peak freshness.

**Fair Trade Practice**

We pay our farmers above market rates to support sustainable agriculture.

Our Harvest

Explore our range of locally sourced products, curated for quality and taste.

[View all products →](#)

Finca El Cafetal

Specialty Coffee

Rich, aromatic beans roasted to perfection. Notes of chocolate and caramel.

[Shop Coffee →](#)

Boyacá Farms

Organic Vegetables

Seasonal greens, root vegetables, and herbs harvested daily.

[Shop Vegetables →](#)



Best Value

The Weekly Harvest Box

Get a curated selection of the season's best produce delivered to your doorstep every week. Customize your box with add-ons like coffee, dairy, and artisanal sauces.

- ★ Save up to 20% compared to individual items
- ★ Reduce food waste with planned harvests
- ★ Pause or cancel anytime
- ★ Free delivery on all subscription boxes

Build Your Box

Our Mission

To revolutionize the Colombian grocery market by building a transparent, sustainable bridge between local farmers and urban households.

Taste the Difference



Community First

We believe in fair trade. We pay our farmers above market rates, ensuring their businesses thrive alongside ours.



Sustainable Farming

We only partner with farms that practice responsible agriculture, protecting our soil and water for future generations.



Innovation

Using advanced forecasting algorithms, we minimize food waste by harvesting exactly what our subscribers need.



Local Roots

We are proudly Colombian. From the mountains of Antioquia to the valleys of Boyacá, we celebrate our land's bounty.

Source: Casewriters

Exhibit 4 Cosecha Fresca Financials from 2020 to 2025

(All figures in COP Billions)	2020	2021	2022	2023	2024	2025 (Proj)
Total Revenue	\$48.50	\$58.20	\$67.00	\$75.80	\$91.40	\$110.00
YoY Growth %	--	20%	15%	13%	21%	20%
Cost of Goods Sold (COGS)	-37.8	-45.4	-52.3	-59.1	-70.4	-84.7
Gross Profit	\$10.70	\$12.80	\$14.70	\$16.70	\$21.00	\$25.30
Gross Margin %	22.00%	22.00%	21.90%	22.00%	23.00%	23.00%
Operating Expenses						
Logistics & Fulfillment	-6.8	-8.1	-9.5	-10.6	-12.5	-14.3
Tech & Development	-1.5	-1.8	-2	-2.3	-3.2	-3.5
Sales & Marketing	-2	-2.3	-2.5	-2.6	-3	-3.3
G&A (Admin/Rent)	-1	-1	-1.1	-1.1	-1.2	-1.3
Total OpEx	-11.3	-13.2	-15.1	-16.6	-19.9	-22.4
EBITDA	-0.6	-0.4	-0.4	\$0.10	\$1.10	\$2.90
Depreciation & Amortization	-0.2	-0.2	-0.3	-0.3	-0.4	-0.5
EBIT (Operating Income)	-0.8	-0.6	-0.7	-0.2	\$0.70	\$2.40
Interest & Tax	-0.1	-0.1	-0.1	0	-0.2	-0.8
Net Income	-0.9	-0.7	-0.8	-0.2	\$0.50	\$1.60
Net Margin %	-1.90%	-1.20%	-1.20%	-0.30%	0.50%	1.50%

Source: Casewriters

Exhibit 5 Cosecha Fresca Top 30 Most Popular Product List

Rank	Product Name	Category	Unit Price (COP)	Net Margin	Net Profit (COP)
1	Finca El Cafetal Ground Medium 500g	Coffee	\$35,000	2.50%	\$875
2	Free-Range Eggs (A) 12-Pack	Dairy/Eggs	\$12,500	1.00%	\$125
3	Premium Dark Roast 340g	Coffee	\$28,000	2.50%	\$700
4	Hass Avocado 1kg	Fruit	\$9,500	2.00%	\$190
5	Farm Fresh Whole Milk (1L)	Dairy	\$5,800	1.00%	\$58
6	Finca El Cafetal Specialty Beans	Coffee	\$38,500	3.00%	\$1,155
7	Tomatoes (Chonto) 1kg	Vegetable	\$5,200	1.50%	\$78
8	Artisanal Masarepa 500g	Pantry	\$5,200	2.00%	\$104
9	Red Onions 1kg	Vegetable	\$4,500	1.50%	\$68
10	Papa Pastusa (White Potato) 1kg	Vegetable	\$4,800	1.20%	\$58
11	Chocolate de Mesa 500g	Pantry	\$10,500	2.00%	\$210
12	Campesino Cheese 500g	Dairy	\$14,500	1.50%	\$218
13	Fresh Cilantro (Bunch)	Vegetable	\$2,500	1.50%	\$38
14	Papa Criolla (Yellow Potato) 1kg	Vegetable	\$6,500	1.20%	\$78
15	Panela (Square Block) 500g	Pantry	\$5,500	2.00%	\$110
16	Mango Tommy 1kg	Fruit	\$6,800	2.00%	\$136
17	Organic Carrots 1kg	Vegetable	\$4,200	1.50%	\$63
18	Lettuce (Batavia) Head	Vegetable	\$4,800	1.50%	\$72
19	Double Cream Cheese 500g	Dairy	\$16,000	1.50%	\$240
20	Papaya Unit	Fruit	\$8,000	2.00%	\$160
21	Farm Butter 250g	Dairy	\$12,000	1.50%	\$180
22	Organic Spinach Bunch	Vegetable	\$5,500	1.50%	\$83
23	Bocadillo Veleño 12-Pack	Pantry	\$7,000	2.50%	\$175
24	Maracuyá (Passion Fruit) 1kg	Fruit	\$6,200	2.00%	\$124
25	Beets 1kg	Vegetable	\$4,000	1.50%	\$60
26	Lulo 1kg	Fruit	\$7,500	2.00%	\$150
27	Artisanal Greek Yogurt 1L	Dairy	\$18,500	1.50%	\$278

Rank	Product Name	Category	Unit Price (COP)	Net Margin	Net Profit (COP)
28	Uchuvas (Golden Berries) 250g	Fruit	\$6,500	2.00%	\$130
29	Organic Kale Bunch	Vegetable	\$6,000	1.50%	\$90
30	Artisanal Honey 300ml	Pantry	\$15,000	2.50%	\$375

Source: Casewriters

Exhibit 6 A Comparison Between the Main ML Methods

Feature	Supervised Learning	Unsupervised Learning	Reinforcement Learning	Self-Supervised Learning (GenAI)
Primary Goal	Predict & Classify: Map inputs to a specific, known output.	Discover Structure: Find hidden patterns or groupings in data.	Optimize Action: Maximize a cumulative reward through trial and error.	Learn Representations: Understand language/context to generate new content.
Training Mechanism	Mapping Inputs to Targets: Uses labeled data to identify the mathematical relationships between input features (e.g., house size) and target outcomes (e.g., price).	Pattern Inference: Analyzes raw, unlabeled data to infer intrinsic structures, densities, or similarities without guidance on what those structures are.	Trial and Error: Interacts with an environment, taking actions and receiving feedback (rewards/penalties) to learn a policy that maximizes long-term success.	Masking & Prediction: Hides parts of the input data (e.g., masking a word in a sentence) and forces the model to predict the missing piece using the surrounding context.
The "Core Question"	Regression: "How much?" (Quantity) Classification: "Which one?" (Category)	"Which of these items are similar to each other?"	"What action should I take next to win?"	"Based on the previous words, what comes next?"
Data Requirements	Labeled Data: Requires input data paired with correct answers (The "Gold Standard").	Unlabeled Data: Raw data without tags or categories.	Environment: A simulation or real-world scenario where the agent can act.	Massive Unlabeled Data: Vast corpus of text/images (labels are generated from the data itself).
The "Feedback Signal"	Error Minimization: The difference between the model's guess and the provided label.	Internal Density: How compact and separated the discovered clusters are.	Reward/Penalty: A positive or negative signal received after taking an action.	Reconstruction: Did the model correctly predict the masked/hidden part of the input?

Feature	Supervised Learning	Unsupervised Learning	Reinforcement Learning	Self-Supervised Learning (GenAI)
Representative Algorithms	• Linear Regression (Predicting values) • Logistic Regression (Predicting probability) • Random Forest / Decision Trees	• K-Means Clustering • Principal Component Analysis (PCA) • Anomaly Detection Algorithms	• Q-Learning • Monte Carlo Tree Search • Deep Q-Network (DQN)	• Transformers (e.g., GPT, BERT) • Diffusion Models (e.g., Stable Diffusion)
Business Applications	• Reg: Demand forecasting, Real estate pricing. • Class: Spam detection, Churn prediction, Image recognition.	• Customer segmentation • Fraud detection (spotting outliers) • Market basket analysis	• Dynamic pricing • Robotics/Warehouse automation • Algorithmic trading	• Content creation (Marketing copy, Images) • Code generation • Semantic search
Strategic Limitation	High Cost: Labeling data is expensive, slow, and prone to human bias.	Interpretability: It is often unclear why the model grouped things a certain way.	Sample Efficiency: Requires millions of “failures” to learn; dangerous in real-world physical ops.	Hallucination: Can confidently generate plausible but factually incorrect information.

Source: Casewriters

Exhibit 7 Selected Supervised Learning Algorithms

Model	Simple Explanation	Best For	Strengths
Decision Trees	Rule-based (if-then) classification	Business explainability	Easy to explain
Random Forest	Ensemble of decision trees with majority rules	Robust classification	Low overfitting
Gradient Boosting (XGBoost, LightGBM, CatBoost)	Sequential tree boosting ranks feature by relevance	High accuracy, personalization	Industry standard
Logistic Regression	Predicts probabilities	Known segments; simple ranking	Interpretable; easy debugging
Linear Regression	Feature-based demand prediction	Baseline forecasting	Transparent
Ridge / Lasso Regression	Regularized linear models	Many features	Stable
Random Forest Regression	Nonlinear regression	Complex demand	Robust
Gradient Boosting Regression	Boosted trees	High accuracy	Industry standard
k-Nearest Neighbors Regression	Uses similar historical periods selecting majority of 'k' neighbors' category	Repetitive patterns	Simple
Support Vector Machines (SVM)	Finds optimal separating margin	High-dimensional data for classification	Strong theory
Naive Bayes	Assume feature independence with Bayes' theorem	Classification of unstructured data	Stable

Model	Simple Explanation	Best For	Strengths
Neural Networks (incl Multi-Layer Perceptron, Convolutional, etc.)	Learns nonlinear decision boundaries	Large platforms, Complex behavior	Flexible, expressive
Autoregressive Integrated Moving Average (ARIMA)	Autoregressive time-series forecasting that predicts future values based on past, univariate data patterns	Stationary demand	Interpretable
Support Vector Regression	Margin-based regression, find ideal boundaries	Small datasets, predicting continuous variables	Stable

Source: Casewriters

Exhibit 8 Selected Unsupervised Learning Algorithms

Model	Simple Explanation	Best For	Strengths
K-Means	Groups customers by similarity into K clusters	Initial segmentation	Fast, simple
Hierarchical Clustering	Builds nested clusters	Exploratory analysis	Interpretable
Gaussian Mixture Models (GMM)	Probabilistic clustering	Soft segmentation	Membership probabilities
User-based Collaborative Filtering	Recommends items liked by similar users	Small user sets	Intuitive
Anomaly Detection	Flags unusual demand	Shock detection	Early warning
Change Point Detection	Detects regime shifts	Trend breaks	Interpretable
Spectral Clustering	Graph-based clustering	Complex shapes	Captures nonlinearity

Source: Casewriters

Exhibit 9 Selected Hybrid Learning Algorithms

Model	Simple Explanation	Best For	Strengths
Cluster → Classify	Uses clusters as labels	Scaling segmentation	Operationally efficient
Self-supervised Segmentation	Learns structure without labels	Cold-start users	Label-free
Global Forecasting Models	Single model across products	Large catalogs	Data-efficient
Wide & Deep Models	Combines memorization + generalization	Mixed features	Proven at scale
Neural Collaborative Filtering	Nonlinear CF	Complex preferences	Better than MF
Graph Neural Networks	Learns over interaction graphs	Rich relationships	Network effects
Sequence Models (LSTM, GRU)	Models purchase order	Repeat buying	Temporal awareness
Transformers (Recommenders)	Attention over history	Long sequences	State-of-the-art

Source: Casewriters

Exhibit 10 Glossary of Key Data Science Terms

Term	Meaning
Algorithm vs. Model	An algorithm is the specific procedure or set of rules used to solve a problem (e.g., “Linear Regression”). A model is the specific artifact created after the algorithm has been trained on data. In business terms, the algorithm is the recipe; the model is the meal prepared for the customer.
Structured Data	Information that is highly organized and formatted, typically in rows and columns (e.g., a customer database or Excel spreadsheet). It is easily searchable and processed by standard algorithms.
Unstructured Data	Information that lacks a pre-defined data model or organization, such as emails, text documents, images, and audio. This data typically requires more complex processing techniques to extract value.
Feature	An individual measurable property or characteristic used as input for a model. For example, in a real estate model, “Square Footage,” “Number of Bedrooms,” and “Zip Code” are features.
Label (or Target)	The outcome or “answer” the model is trying to predict. In a dataset used to train a churn model, the label would be the column indicating whether the customer actually cancelled their subscription.
Training	The phase where the algorithm analyzes the dataset to “learn” the underlying patterns and relationships. This is often computationally expensive and time-consuming.
Inference	The phase where the trained model is deployed to make predictions on new, live data. This is the operational stage where the model delivers value.
Train/Test Split	The practice of splitting the available dataset into two parts: a training set used to build the model, and a test set (or holdout set) used to evaluate its performance. This ensures the model is tested on data it has never seen before.
Overfitting	A modeling error where a function is too closely aligned to a limited set of data points. An overfitted model has memorized the noise and details of the training data rather than the general trend, leading to poor performance on new data.

Term	Meaning
Underfitting	A scenario where the model is too simple to capture the underlying structure of the data (e.g., trying to use a straight line to model a curve). Unlike overfitting, an underfitted model performs poorly on <i>both</i> the training data and the live data.
False Positive (Type I Error)	An error where the model incorrectly predicts the positive class. For example, flagging a legitimate transaction as fraud (a “false alarm”).
False Negative (Type II Error)	An error where the model incorrectly predicts the negative class. For example, failing to identify a fraudulent transaction (a “missed detection”).
Accuracy	The ratio of correctly predicted observations to the total observations. Note: This can be misleading in unbalanced datasets (e.g., a fraud model could be 99.9% accurate by simply predicting “no fraud” every time).

Source: Casewriters

Endnotes

¹ Bojinov, I. (2023). Keep your AI projects on track. *Harvard Business Review*, 11, 2023.